

Mechanical Engineering Experiments (II) — Report 1

機械工程實驗(二) 第一次報告

Student ID / 學號 : B12502027 Class / 班次 : 02 Name / 姓名 : 李承軒

Q1:

(a) .

Upstream velocity (m/s):

Table 1. Height – Velocity Chart under different duties (upstream)

		15% duty	17% duty	20% duty
Height	4 cm	2.03	2.77	2.67
	8 cm	2.21	2.84	3.14
	12 cm	2.98	2.42	4.24
	16 cm	2.75	5.55	6.74
	20 cm	6.45	7.38	8.4
	24 cm	6.82	7.76	9.07
	28 cm	6.86	7.97	9.29

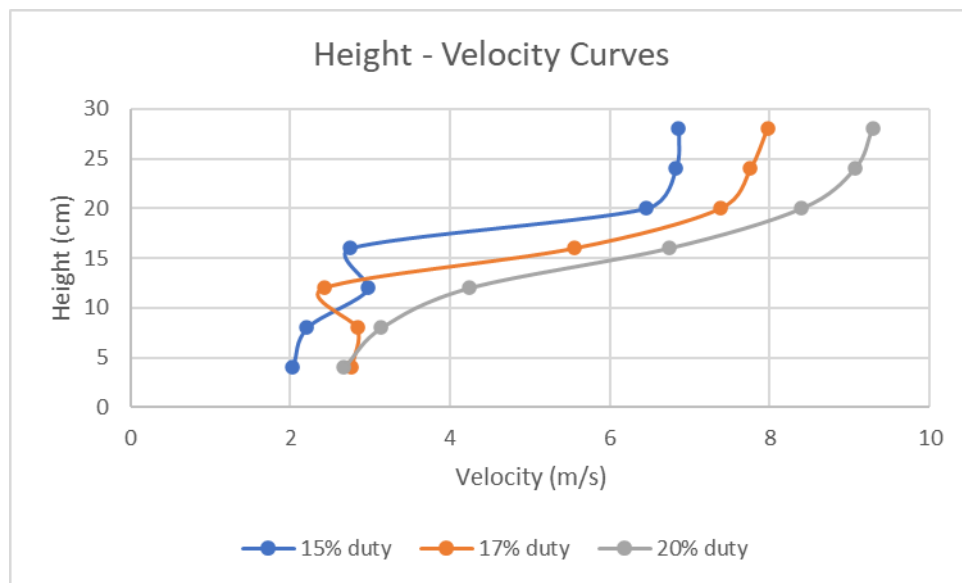


Figure 1. Height – Velocity Curves under different duties (upstream)

Downstream velocity (m/s):

Table 2. Height – Velocity Chart under different duties (downstream)

		15% duty	17% duty	20% duty
Height	4 cm	0.91	0.78	0.83
	8 cm	0.71	0.81	0.91
	12 cm	0.78	0.77	0.8
	16 cm	1.35	1.53	1.57
	20 cm	2.2	2.18	2.71
	24 cm	8.01	8.94	10.15
	28 cm	8.13	9.5	11.1

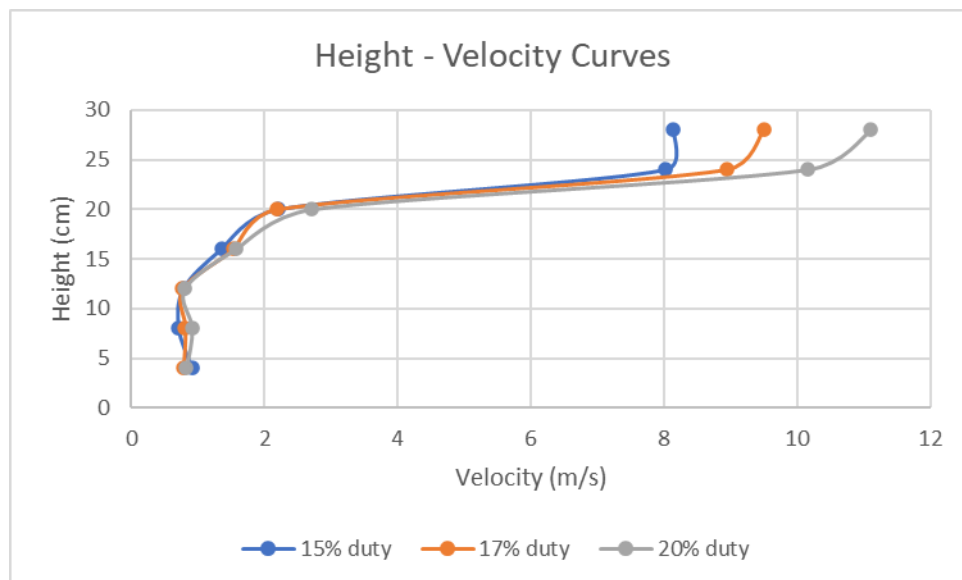


Figure 2. Height – Velocity Curves under different duties (downstream)

(b) .

$$\rho_{air @ room temp.} = 1.293 \text{ kg/m}^3$$

$$\Delta A = box width \times 0.04 = 0.2225 \times 0.04 = 0.0089 \text{ m}^2$$

For 15% duty:

Table 3. Total Drag Force under 15% duty

u_out (m/s)	u_in (m/s)	(u_out^2 - u_in^2)*dA*density_air (N)
0.91	2.03	-0.037892555
0.71	2.21	-0.050403726
0.78	2.98	-0.095191694
1.35	2.75	-0.066054198
2.2	6.45	-0.423051821
8.01	6.82	0.203084437
8.13	6.86	0.219075537
Total Drag Force		-0.25043402

$$\text{Total Drag Force} = -0.25043402 \text{ N}$$

For 17% duty:

Table 4. Total Drag Force under 17% duty

u_out (m/s)	u_in (m/s)	(u_out^2 - u_in^2)*dA*density_air (N)
0.78	2.77	-0.081296147
0.81	2.84	-0.085266303
0.77	2.42	-0.060570779
1.53	5.55	-0.327527554
2.18	7.38	-0.572070782
8.94	7.76	0.226770736
9.5	7.97	0.307590464
Total Drag Force		-0.592370365

$$\text{Total Drag Force} = -0.592370365 \text{ N}$$

For 20% duty:

Table 5. Total Drag Force under 20% duty

u_out (m/s)	u_in (m/s)	(u_out^2 - u_in^2)*dA*density_air (N)
0.83	2.67	-0.074109588
0.91	3.14	-0.103931793
0.8	4.24	-0.1995159
1.57	6.74	-0.494401863
2.71	8.4	-0.727469612
10.15	9.07	0.238872234
11.1	9.29	0.424702025
Total Drag Force		-0.935854496

Total Drag Force = -0.935854496 N

Q2:

(a) .

1. Lift to AOA

Table 6. Lift_true to AOA Chart (fan duty = 15%)

AOA (deg)	Lift_measured (N)	Drag_measured (N)	Lift_true (N)
0	0.203789	0.047329	0.203789
1.5	0.264441	0.048433	0.263172737
3	0.366142	0.050471	0.363496927
4.4	0.442231	0.053721	0.438097394
5.9	0.538509	0.056509	0.532669363
7.5	0.607996	0.059688	0.600137926
9.1	0.690336	0.064445	0.680013583
10.6	0.809302	0.071422	0.795935736
12.1	0.869739	0.083736	0.851787557
13.5	0.904536	0.094873	0.881759008
14.6	0.946106	0.105017	0.91875112
15.9	0.948134	0.107858	0.91740984
17.4	0.850371	0.115078	0.814307741
19.1	0.861377	0.125539	0.817905224
20.6	0.771173	0.131956	0.721574

Table 7. Lift_true to AOA Chart (fan duty = 20%)

AOA (deg)	Lift_measured (N)	Drag_measured (N)	Lift_true (N)
0	0.379745	0.086731	0.379745
1.5	0.523849	0.089604	0.521502637
3	0.677233	0.093608	0.672327213
4.4	0.794765	0.098447	0.787189916
5.9	0.856216	0.099725	0.845910424
7.5	1.096363	0.10842	1.082089236
9.1	1.17456	0.116552	1.155891394
10.6	1.40456	0.124418	1.381275804
12.1	1.558702	0.1449	1.527638132
13.5	1.631919	0.164352	1.592461576
14.6	1.794185	0.180781	1.747095077
15.9	1.818084	0.1943889	1.762710865
17.4	1.9103	0.21527	1.842838471
19.1	1.855961	0.225608	1.777837225
20.6	1.53127	0.234825	1.443005085

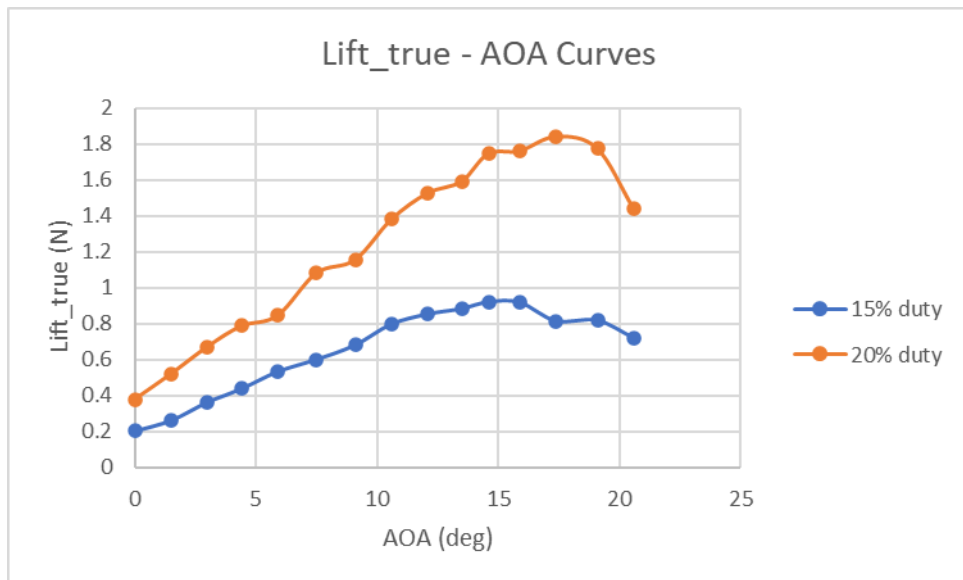


Figure 3. Lift_true - AOA Curves under different duties

2. Drag to AOA

Table 8. Drag to AOA Chart (fan duty = 15%)

AOA (deg)	Drag_measured (N)
0	0.047329
1.5	0.048433
3	0.050471
4.4	0.053721
5.9	0.056509
7.5	0.059688
9.1	0.064445
10.6	0.071422
12.1	0.083736
13.5	0.094873
14.6	0.105017
15.9	0.107858
17.4	0.115078
19.1	0.125539
20.6	0.131956

Table 9. Drag to AOA Chart (fan duty = 20%)

AOA (deg)	Drag_measured (N)
0	0.086731
1.5	0.089604
3	0.093608
4.4	0.098447
5.9	0.099725
7.5	0.10842
9.1	0.116552
10.6	0.124418
12.1	0.1449
13.5	0.164352
14.6	0.180781
15.9	0.1943889
17.4	0.21527
19.1	0.225608
20.6	0.234825

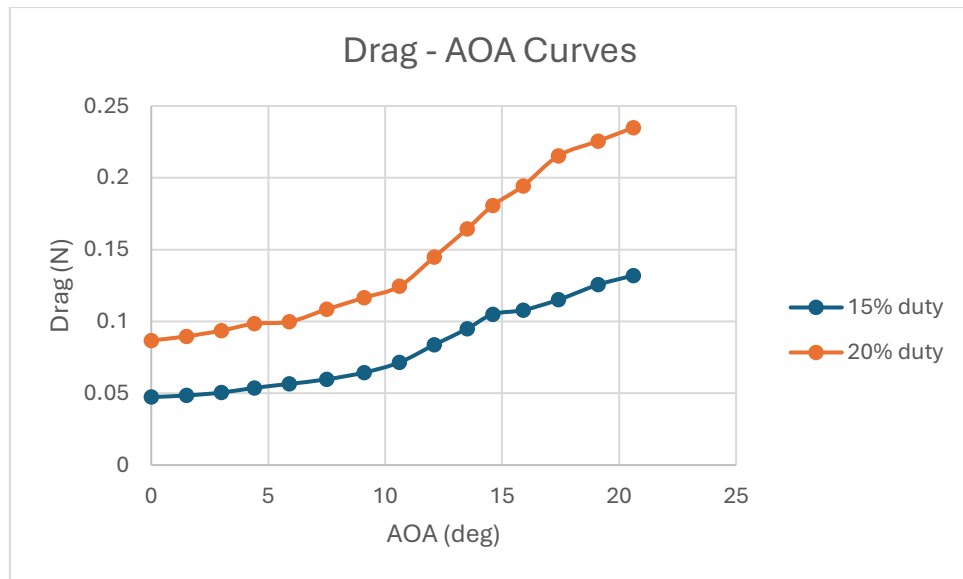


Figure 4. Drag - AOA Curves under different duties

3. Lift Coefficient to AOA

$$\text{Lift Coefficient}(C_L) = \frac{L}{0.5\rho u^2 A}$$

$$\rho = 1.293 \text{ kg/m}^3 \quad A = 0.28 \times 0.21 = 0.0588 \text{ m}^2$$

Table 10. Lift coefficient to AOA Chart (fan duty = 15%)

AOA (deg)	wind speed u (m/s)	Lift true (N)	Lift Coefficient
0	6.77	0.203789	0.116965436
1.5	6.95	0.263172737	0.143326149
3	7.22	0.363496927	0.183434277
4.4	6.84	0.438097394	0.246327337
5.9	6.79	0.532669363	0.303929121
7.5	6.79	0.600137926	0.342425161
9.1	6.83	0.680013583	0.383469057
10.6	6.96	0.795935736	0.432228769
12.1	7.04	0.851787557	0.452105841
13.5	6.97	0.881759008	0.477461649
14.6	6.99	0.91875112	0.494649614
15.9	6.82	0.91740984	0.518858325
17.4	6.86	0.814307741	0.455191849
19.1	6.8	0.817905224	0.465306695
20.6	6.87	0.721574	0.402180998

Table 11. Lift coefficient to AOA Chart (fan duty = 20%)

AOA (deg)	wind speed u (m/s)	Lift _{true} (N)	Lift Coefficient
0	9.48	0.379745	0.111155136
1.5	9.48	0.521502637	0.152649005
3	9.47	0.672327213	0.197212699
4.4	9.5	0.787189916	0.229449152
5.9	9.48	0.845910424	0.247606389
7.5	9.47	1.082089236	0.317407558
9.1	9.46	1.155891394	0.339773029
10.6	9.45	1.381275804	0.4068843
12.1	9.3	1.527638132	0.464631594
13.5	9.52	1.592461576	0.462220507
14.6	9.6	1.747095077	0.498687197
15.9	9.39	1.762710865	0.525901055
17.4	9.37	1.842838471	0.552156556
19.1	9.4	1.777837225	0.529286032
20.6	9.52	1.443005085	0.41883996

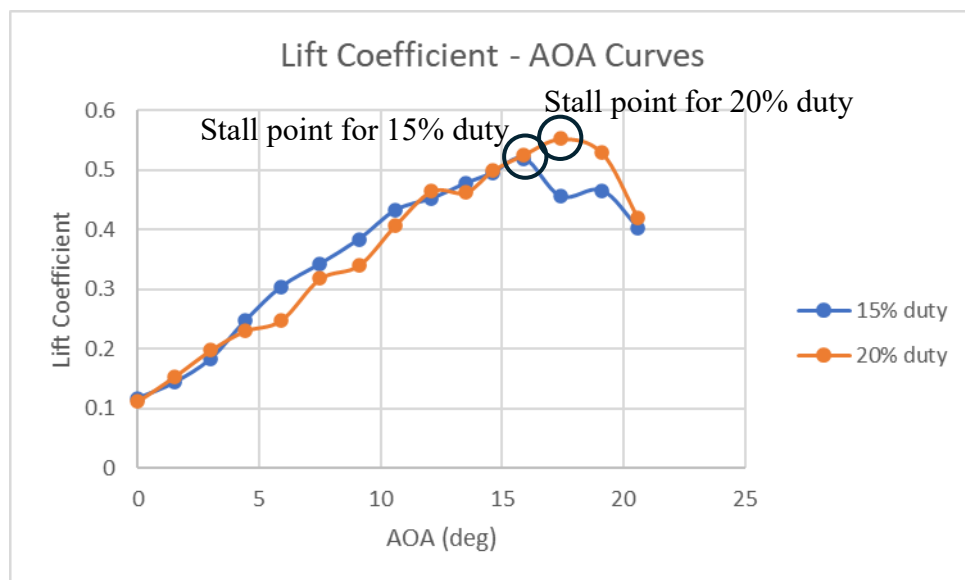


Figure 5. Lift Coefficient - AOA Curves under different duties

For 15% duty: Stall point is at AOA $\approx 15.9^\circ$

For 20% duty: Stall point is at AOA $\approx 17.4^\circ$

4. Drag coefficient to AOA

$$\text{Drag Coefficient}(C_d) = \frac{D}{0.5\rho u^2 A}$$

$$\rho = 1.293 \text{ kg/m}^3 \quad A = 0.28 \times 0.21 = 0.0588 \text{ m}^2$$

Table 12. Drag coefficient to AOA Chart (fan duty = 15%)

AOA (deg)	wind speed u (m/s)	Drag_measured (N)	Drag Coefficient
0	6.77	0.047329	0.027164651
1.5	6.95	0.048433	0.026377031
3	7.22	0.050471	0.025469573
4.4	6.84	0.053721	0.0302055
5.9	6.79	0.056509	0.032242761
7.5	6.79	0.059688	0.034056626
9.1	6.83	0.064445	0.036341426
10.6	6.96	0.071422	0.038785346
12.1	7.04	0.083736	0.044444808
13.5	6.97	0.094873	0.051372562
14.6	6.99	0.105017	0.056540468
15.9	6.82	0.107858	0.061001113
17.4	6.86	0.115078	0.064327729
19.1	6.8	0.125539	0.0714192
20.6	6.87	0.131956	0.073547821

Table 13. Drag coefficient to AOA Chart (fan duty = 20%)

AOA (deg)	wind speed u (m/s)	Drag_measured (N)	Drag Coefficient
0	9.48	0.086731	0.025387026
1.5	9.48	0.089604	0.026227981
3	9.47	0.093608	0.02745789
4.4	9.5	0.098447	0.028695211
5.9	9.48	0.099725	0.029190499
7.5	9.47	0.10842	0.031802671
9.1	9.46	0.116552	0.034260335
10.6	9.45	0.124418	0.03664998
12.1	9.3	0.1449	0.044071378
13.5	9.52	0.164352	0.047704049
14.6	9.6	0.180781	0.051601754
15.9	9.39	0.1943889	0.057995517
17.4	9.37	0.21527	0.064499816
19.1	9.4	0.225608	0.067166533
20.6	9.52	0.234825	0.068159215

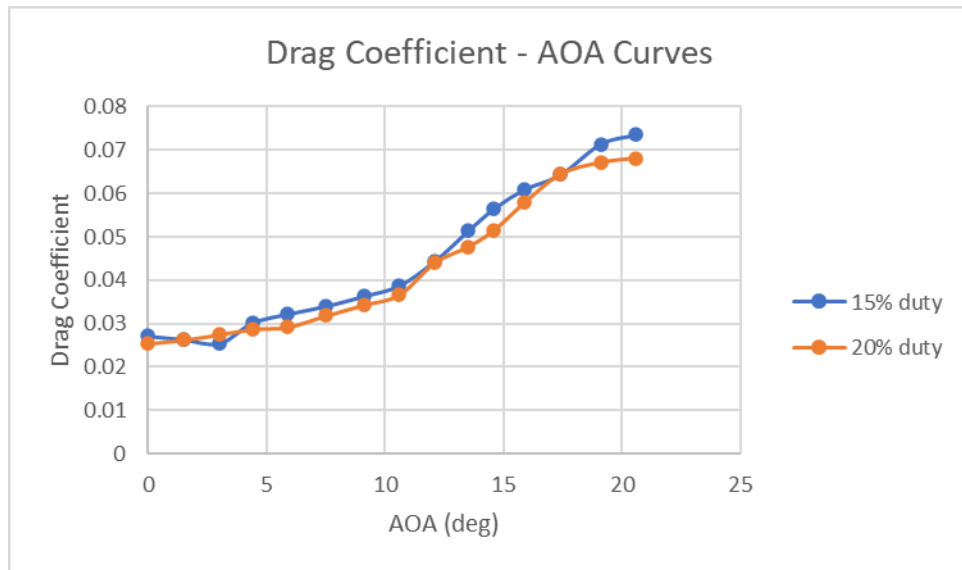


Figure 6. Drag Coefficient - AOA Curves under different duties

5. Ratio of L/D to AOA

Table 14. L/D to AOA Chart (fan duty = 15%)

AOA (deg)	Lift_true (N)	Drag_measured (N)	L/D
0	0.203789	0.047329	4.305795601
1.5	0.263172737	0.048433	5.433748421
3	0.363496927	0.050471	7.202094806
4.4	0.438097394	0.053721	8.15504913
5.9	0.532669363	0.056509	9.426274804
7.5	0.600137926	0.059688	10.05458259
9.1	0.680013583	0.064445	10.55184395
10.6	0.795935736	0.071422	11.14412556
12.1	0.851787557	0.083736	10.17229814
13.5	0.881759008	0.094873	9.294098509
14.6	0.91875112	0.105017	8.748594232
15.9	0.91740984	0.107858	8.505719005
17.4	0.814307741	0.115078	7.07613741
19.1	0.817905224	0.125539	6.515148469
20.6	0.721574	0.131956	5.468292464

Table 15. L/D to AOA Chart (fan duty = 20%)

AOA (deg)	Lift_true (N)	Drag_measured (N)	L/D
0	0.379745	0.086731	4.37842294
1.5	0.521502637	0.089604	5.820082102
3	0.672327213	0.093608	7.182369163
4.4	0.787189916	0.098447	7.996078257
5.9	0.845910424	0.099725	8.482430923
7.5	1.082089236	0.10842	9.980531601
9.1	1.155891394	0.116552	9.917387898
10.6	1.381275804	0.124418	11.10189687
12.1	1.527638132	0.1449	10.54270622
13.5	1.592461576	0.164352	9.689334938
14.6	1.747095077	0.180781	9.664152077
15.9	1.762710865	0.1943889	9.067960492
17.4	1.842838471	0.21527	8.560591217
19.1	1.777837225	0.225608	7.880204716
20.6	1.443005085	0.234825	6.145023253

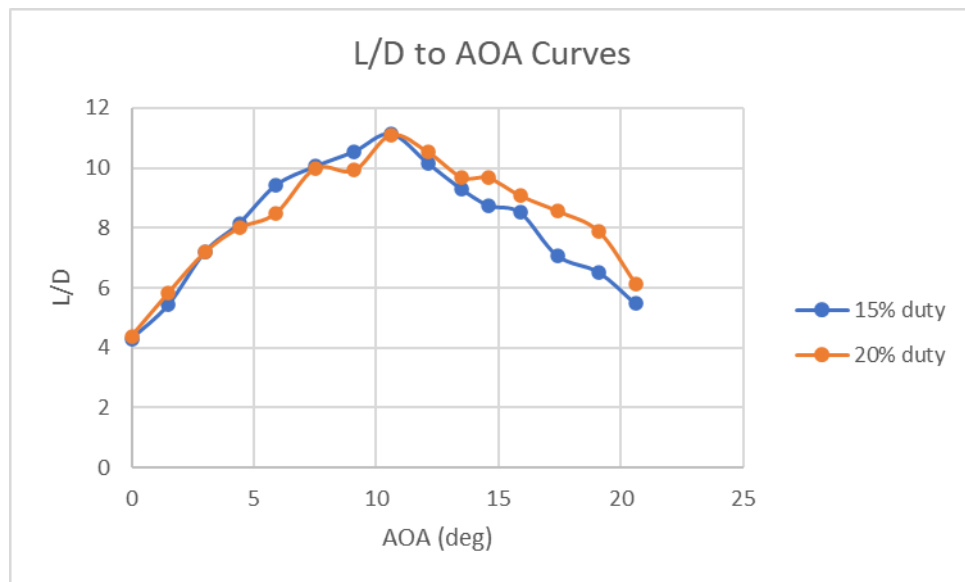


Figure 7. L/D - AOA Curves under different duties

(b) .

Airfoil has the best performance when ratio of L/D reaches its maximum value

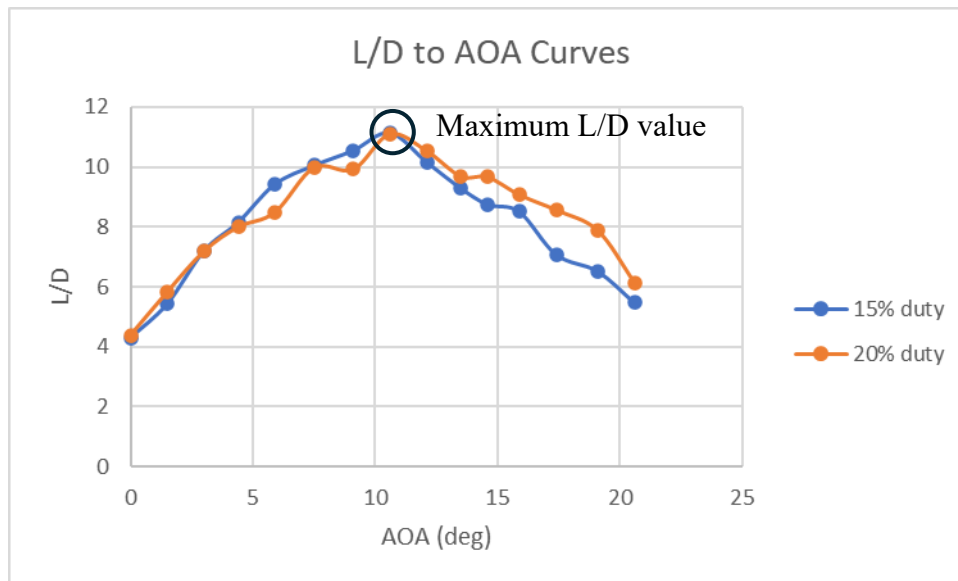


Figure 8. L/D - AOA Curves under different duties

When $AOA = 10.6^\circ$, L/D reaches its maximum value 11.14412556

At this point, wind speed = 6.96 m/s

Q3:

(a) .

(a-1) . Velocity is assumed to follow the uniform distribution across the wind tunnel during the transition between 2D and 3D CV analysis, which means the velocity measured at the midline represents the velocity at the same height across the entire span of the tunnel.

(a-2) . NO, the assumption does not hold when the flow passes the box. The box and the wind tunnel wall create boundary layer effects and three-dimensional flow structures, which cause the velocity to vary across the width of the tunnel.

(b) .

(b-1) . For two points directly on the CV boundary in the steady flow:

$$Total\ Force = \rho(u_{out}^2 dA - u_{in}^2 dA)$$

(b-2) . For Bernoulli Equation $\Delta P = 0.5\rho\Delta(u^2)$:

$$Force = \Delta P \times Area = 0.5\rho(u_{out}^2 dA - u_{in}^2 dA)$$

Compared to the CV result in (b-1), both formulas relate the force to the difference of velocity squares, but the Bernoulli estimation includes a coefficient of 0.5 due to the energy conservation assumption.

(b-3-1) . For segments 1 and 2, the Bernoulli derivation generally holds (with small viscous errors), because the flow is accelerating or decelerating outside the wake region, and the energy conservation assumption is relatively valid.

For segment 3, the Bernoulli derivation does not hold. This segment contains the turbulent region behind the box, where energy dissipates due to vortices.

(b-3-2) . Segment 1 provides a bigger force exerting on CV measured. Because segment 1 is near the stagnation point, it decelerates significantly and creates large pressure buildup, which leads to a bigger force.

(c) .

1. The downstream measurement point should be closer to the box because when the flow move further away, it will mix with the surrounding free stream and the velocity deficit caused by the box drag will dissipate, leading to an underestimation of actual momentum loss.
2. The upstream measurement point at a distance from the box introduces an error where the pressure and velocity have not yet been influenced by the box. This leads to underestimation because the CV fail to capture the full pressure gradient and flow deceleration caused by the box.